

0.3656 rad/s and 2.5456 rad/s, respectively. Comparing these values with those of Example 2.25 with lead compensation, we find that there is a slight increase in the gain margin, while the phase margin has been reduced from infinite to 104.5° . Furthermore, there is a slight slowing down of the closed-loop response due to the added lag compensation, which is evident in a slight reduction in the phase crossover frequency. The slowing down of the closed-loop response is also clear in the overshoot and settling time of the step response. The lead compensated system of Example 2.25 had an 8 percent overshoot and settling time of 19 seconds. To achieve the same overshoot, the lead-lag compensated system would require a settling time slightly larger than 20 seconds, which is obvious from Figure 2.53. Example 2.26 illustrates the iterative process of determining the three unknowns (α , ω_1 , ω_2) of a lead-lag compensator to meet a conflicting set of performance requirements. Since a lead-lag compensator is a practical (approximate) implementation of the ideal PID compensator, we have witnessed an example of PID tuning in which the three unknown constants K_P , K_D , and K_I are determined in a similar manner.

The Control System Toolbox (CST) offers an interactive *Graphical Users Interface* (GUI) tool, called the *SISO Design Tool*, to carry out the steps illustrated in this section for the design of closed-loop compensators in a user-friendly manner. To use this design feature, go to MATLAB *launch pad*, click on the *Control System Toolbox*, and select the icon for *SISO Design Tool* from the resulting menu. Another interactive GUI tool available in the CST menu is the *LTI Viewer*, which lets you directly view all the important graphs of a linear, time-invariant system (step and impulse responses, Bode, Nyquist, root-locus, Nichols plots, and pole-zero map, etc.), as well as the performance characteristics of the system (peak response, rise time, settling time, and steady-state), along with several graphical operations. As an exercise to familiarize yourself with these GUI tools, repeat Examples 2.23–2.26 with the *SISO Design Tool*, and use *LTI Viewer* to check the closed-loop system's characteristics.

2.13 Multivariable Systems

So far we have considered only single-input, single-output systems (an exception was the aircraft in Example 2.10, which had three outputs, and one input). A practical modern control system may simultaneously contain several inputs and several outputs. Such systems are called *multivariable systems*. For multivariable systems, we must re-define the relationship between the Laplace transform of the input, $\mathbf{U}(s)$, and the Laplace transform of the output, $\mathbf{Y}(s)$, both of which are now *vectors*, consisting of more than one signal. Each element of $\mathbf{U}(s)$ is the Laplace transform of the *corresponding element* of the input vector, $\mathbf{u}(t)$. Similarly, each element of $\mathbf{Y}(s)$ is the Laplace transform of the *corresponding element* of the output vector, $\mathbf{y}(t)$. The input and output vectors are related in the Laplace domain by a *transfer matrix* (instead of the transfer function for the single-input, single-output systems), $\mathbf{G}(s)$, which is defined as follows:

$$\mathbf{Y}(s) = \mathbf{G}(s)\mathbf{U}(s) \quad (2.188)$$

Note that we are using *bold letters* to denote *vectors* and *matrices*. This notation will be followed throughout the book. Refer to Appendix B, or Gantmacher [4] and Kreyszig [5] for vectors, matrices, and their properties.

In Eq. (2.188) if the system consists of p outputs and r inputs, then the sizes of $\mathbf{Y}(s)$, $\mathbf{U}(s)$, and $\mathbf{G}(s)$ are $(p \times 1)$, $(r \times 1)$, and $(p \times r)$, respectively. (Clearly, the single-input, single-output system is a special case of the multivariable systems where $p = 1$ and $r = 1$.) Since we are dealing with vectors and matrices here, we have to be careful how we *multiply* them such that the sizes of the multiplied quantities are *compatible* (e.g. we can not write Eq. (2.188) as $\mathbf{Y}(s) = \mathbf{U}(s)\mathbf{G}(s)!$).

Example 2.27

Let us derive the transfer matrix of the multivariable system of Example 2.10. Recall that the system (longitudinal dynamics of an aircraft) consists of a *single* input whose Laplace transform is $\delta(s)$, and *three* outputs with Laplace transforms $v(s)$, $\alpha(s)$, and $\theta(s)$. Thus, we can write

$$\mathbf{U}(s) = \delta(s); \quad \mathbf{Y}(s) = \begin{bmatrix} v(s) \\ \alpha(s) \\ \theta(s) \end{bmatrix} \quad (2.189)$$

The transfer matrix between $\mathbf{G}(s)$ can be obtained using the relationships between the input and the individual elements of $\mathbf{Y}(s)$ given by Eqs. (2.86)–(2.88) as follows:

$$\mathbf{Y}(s) = \begin{bmatrix} v(s) \\ \alpha(s) \\ \theta(s) \end{bmatrix} = \begin{bmatrix} v(s)/\delta(s) \\ \alpha(s)/\delta(s) \\ \theta(s)/\delta(s) \end{bmatrix} \mathbf{U}(s) \quad (2.190)$$

or

$\mathbf{Y}(s)$

$$= \begin{bmatrix} -0.0005(s - 70)(s + 0.5)/[(s^2 + 0.005s + 0.006)(s^2 + s + 1.4)] \\ -0.02(s + 80)(s^2 + 0.0065s + 0.006)/[(s^2 + 0.005s + 0.006)(s^2 + s + 1.4)] \\ -1.4(s + 0.02)(s + 0.4)/[(s^2 + 0.005s + 0.006)(s^2 + s + 1.4)] \end{bmatrix} \mathbf{U}(s) \quad (2.191)$$

which, from Eq. (2.188), gives

$\mathbf{G}(s)$

$$= \begin{bmatrix} -0.0005(s - 70)(s + 0.5)/[(s^2 + 0.005s + 0.006)(s^2 + s + 1.4)] \\ -0.02(s + 80)(s^2 + 0.0065s + 0.006)/[(s^2 + 0.005s + 0.006)(s^2 + s + 1.4)] \\ -1.4(s + 0.02)(s + 0.4)/[(s^2 + 0.005s + 0.006)(s^2 + s + 1.4)] \end{bmatrix} \quad (2.192)$$

Note that the transfer matrix, $\mathbf{G}(s)$, is of size (3×1) , denoting a single-input, three-output system.

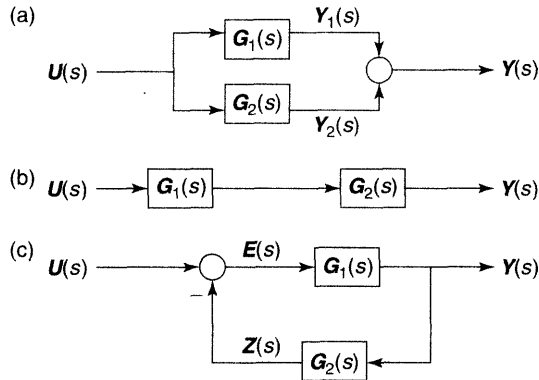


Figure 2.55 Three types of connections in multivariable control systems: (a) parallel, (b) series, and (c) negative feedback

We must be able to determine a system's transfer matrix from the block diagram of a multivariable system, which may include transfer matrices of the various sub-systems. As in single-input, single-output (SISO) systems, there are three distinct ways in which the multivariable sub-systems can be connected to each other, as shown in Figure 2.55. Figure 2.55(a) depicts two *sub-systems* of transfer matrices $\mathbf{G}_1(s)$ and $\mathbf{G}_2(s)$, respectively, connected in *parallel*. Here, the outputs of the two sub-systems, $\mathbf{Y}_1(s)$ and $\mathbf{Y}_2(s)$ (which clearly must be vectors of the *same size*) are added-up at the *summing junction* to give the system's output, $\mathbf{Y}(s)$, by

$$\mathbf{Y}(s) = \mathbf{Y}_1(s) + \mathbf{Y}_2(s) = \mathbf{G}_1(s)\mathbf{U}(s) + \mathbf{G}_2(s)\mathbf{U}(s) = [\mathbf{G}_1(s) + \mathbf{G}_2(s)]\mathbf{U}(s) \quad (2.193)$$

Hence, the parallel system's transfer matrix is the sum of the transfer matrices of its two (or more) components, i.e. $\mathbf{G}(s) = \mathbf{G}_1(s) + \mathbf{G}_2(s)$. The system of Figure 2.55(b) is in *series*, and it can be easily seen (left as an exercise for you) that the transfer matrix of such a system is given by the *product* of the transfer matrices of the two (or more) components, i.e. $\mathbf{G}(s) = \mathbf{G}_2(s)\mathbf{G}_1(s)$. In series combinations, one has to be careful in the *sequence of multiplication* (i.e. $\mathbf{G}_2(s)\mathbf{G}_1(s)$, and *not* $\mathbf{G}_1(s)\mathbf{G}_2(s)$, etc.), because the matrix multiplication is not always *conformable* (i.e. for a matrix product $\mathbf{G}(s) = \mathbf{G}_2(s)\mathbf{G}_1(s)$ to be defined, the number of *columns* in $\mathbf{G}_1(s)$ must be equal to the number of *rows* in $\mathbf{G}_2(s)$) (see Appendix B). The third kind of connection, shown in Figure 2.55(c) is the *feedback* arrangement. The two sub-systems of this combination are the *forward transmission*, given by the transfer matrix $\mathbf{G}_1(s)$, and the *feedback transmission*, whose transfer matrix is $\mathbf{G}_2(s)$. If the *feedback signal*, $\mathbf{Z}(s)$, is subtracted from the input, $\mathbf{U}(s)$ – as in Figure 2.55(c) – then the system is said to be of a *negative feedback* kind, and if $\mathbf{Z}(s)$ is added to $\mathbf{U}(s)$ at the summing junction, the system is called a *positive feedback system*. Again, as in SISO systems, we usually encounter the negative feedback arrangement in multivariable control systems, whose objective is to drive the inverse Laplace transform of the *error vector*, $\mathbf{E}(s) = \mathbf{U}(s) - \mathbf{Z}(s)$, to zero as $t \rightarrow \infty$. For a negative feedback system, we can write

$$\mathbf{Y}(s) = \mathbf{G}_1(s)\mathbf{E}(s) = \mathbf{G}_1(s)[\mathbf{U}(s) - \mathbf{G}_2(s)\mathbf{Y}(s)] \quad (2.194)$$

or

$$[\mathbf{I} + \mathbf{G}_1(s)\mathbf{G}_2(s)]\mathbf{Y}(s) = \mathbf{G}_1(s)\mathbf{U}(s) \quad (2.195)$$

and it follows that

$$\mathbf{Y}(s) = \mathbf{G}(s)\mathbf{U}(s) = [\mathbf{I} + \mathbf{G}_1(s)\mathbf{G}_2(s)]^{-1}\mathbf{G}_1(s)\mathbf{U}(s) \quad (2.196)$$

where $[\]^{-1}$ denotes the *inverse* of a (square) matrix, and $\mathbf{I}$ is the *identity matrix* of the same size as the product $\mathbf{G}_1(s)\mathbf{G}_2(s)$ (Appendix B). Hence, from Eq. (2.196), the transfer matrix of a negative feedback system is given by $\mathbf{G}(s) = [\mathbf{I} + \mathbf{G}_1(s)\mathbf{G}_2(s)]^{-1}\mathbf{G}_1(s)$. (Note that for a SISO system, the transfer matrix (function) becomes $G(s) = G_1(s)/[1 + G_1(s)G_2(s)]$.) Again, we must be careful with the sequence of matrix multiplication involved in expressions such as Eq. (2.196). Not all multivariable closed-loop systems are of the kind shown in Figure 2.55(c).

Trying to find the transfer matrices of multivariable systems from their block diagrams can be a daunting task. The Control System Toolbox (CST) helps in constructing the transfer function from its individual components with model building commands, such as *series*, *parallel*, *feedback*, and *append*. For example, the multivariable system of Figure 2.55(a) can be constructed using the command *parallel* as follows:

```
>>sys = parallel (G1,G2) <enter>
```

where G_1 , G_2 are the LTI objects (either transfer matrices, or *state-space* models (see Chapter 3)) of the two parallel sub-systems, $\mathbf{G}_1$ and $\mathbf{G}_2$, respectively. Alternatively, we could type

```
>>sys = G1+G2 <enter>
```

to get the transfer matrix (or *state-space* model) of the overall system. Similarly, we can use the *series* command for the system of Figure 2.55(b) as follows:

```
>>sys = series(G1,G2) <enter>
```

or type

```
>>sys = G2*G1 <enter>
```

For multivariable systems in which only *some* of the outputs of G_1 are the inputs to G_2 , both *parallel* and *series* commands can be used with *two additional input arguments* such as

```
>>sys = series(G1,G2,OUTPUTS1,INPUTS2) <enter>
```

which specifies that the outputs of G_1 denoted by the *index vector* OUTPUTS1 are connected to the inputs of G_2 denoted by INPUTS2. (The vectors OUTPUTS1 and INPUTS2 contain indices into the outputs and inputs of G_1 and G_2 , respectively.) For the

feedback multivariable system the CST command *feedback* gives the closed-loop transfer matrix as follows:

```
>>sys = feedback(G1,G2,FEEDIN,FEEDOUT,sign) <enter>
```

The vector FEEDIN contains indices into the input vector of G1, and specifies which inputs are involved in the feedback loop. Similarly, FEEDOUT specifies which outputs of G1 are used for feedback. If the sign is omitted, a negative feedback system is assumed, while sign=1 results in a positive feedback system. If FEEDIN and FEEDOUT are omitted from the feedback command, it is assumed that all the outputs of G1 are the inputs to G2 (as shown in Figure 2.55(c)).

Example 2.28

Consider the multivariable control system of Figure 2.56. The system's block-diagram indicates that only the *second* and *third* outputs of $G_1(s)$ are inputs to $G_2(s)$, while the *second* output of $G_2(s)$ is an input to $G_3(s)$. It is also clear that $G_1(s)$ and $G_2(s)$ are in series, while $G_3(s)$ is in a negative feedback arrangement with $G_1(s)$ and $G_2(s)$. The closed-loop system's transfer matrix from the input vector, $U(s) = U_1(s)$, and the output vector, $Y(s) = [Y_2(s) Y_3(s)]^T$ can be determined using the CST commands as follows:

```
>>G1=tf([1; -2; 1],[1 1]; [1 0]; [1 0 5]) % transfer matrix,G1 <enter>
```

Transfer function from input to output...

```

      1
#1:  ----
      s+1

      -2
#2:  --
      s

```

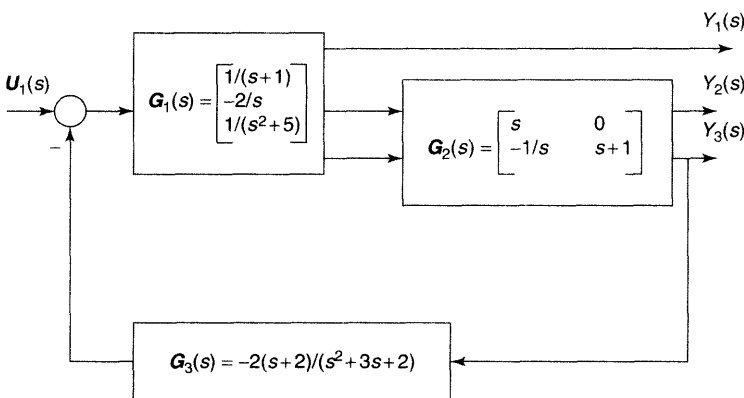


Figure 2.56 A multivariable closed-loop system consisting of sub-systems, G_1 , G_2 , and G_3

```

      1
#3:  ----
      s^2+5

```

```
>>G2=tf([1 0] 0;-1 [1 1]),[1 1];[1 0] 1)) % transfer matrix, G2 <enter>
```

Transfer function from input 1 to output...

```
#1: s
```

```

      -1
#2:  --
      s

```

Transfer function from input 2 to output...

```
#1: 0
```

```
#2: s+1
```

```
>>G3=tf(-2*[1 2],[1 3 2]) % transfer matrix, G3 <enter>
```

Transfer function:

```

      -2s-4
      -----
      s^2+3s+2

```

(Note the ease with which the transfer matrices are constructed using the LTI object, *tf*. The coefficients of the numerator and denominator polynomials are specified using `{·}`, `{·}`, with the transfer functions on different rows separated by`;`.)

```
>>G2G1=series(G1,G2,[2 3],[1 2]) %transfer matrix of the series
connection <enter>
```

Transfer function from input to output...

```

      -2s
#1:  ----
      s

      s^3+3s^2+10
#2:  -----
      s^4+5s^2

```

```
>>GCL=feedback(G2G1, G3, 1, 2) % closed-loop transfer matrix from U1(s)
to [Y2(s); Y3(s)] <enter>
```

Transfer function from input to output...

```

      -2s^6-6s^5-14s^4-30s^3-20s^2+7.43e-015s
#1:  -----
      s^6+3s^5+5s^4+5s^3-2s^2-20s-40

```

$$\#2: \frac{s^5 + 6s^4 + 11s^3 + 16s^2 + 30s + 20}{s^6 + 3s^5 + 5s^4 + 5s^3 - 2s^2 - 20s - 40}$$

Thus the closed-loop transfer matrix indicates a sixth order, single-input, two-output system. The transfer function of the forward path from $U(s) = U_1(s)$ to $Y_1(s)$ is easily seen from Figure 2.56 to be $1/(s + 1)$. Note that the denominator polynomials of the two transfer functions constituting GCL are the same, which indicates that the closed-loop system's poles can be obtained by factoring the common denominator polynomial. The closed-loop zeros are the zeros of transfer functions constituting GCL , which are easily identified from Figure 2.56 as $z_1 = -1$ and $z_2 = -2$. Let us determine the closed-loop zeros and poles using CST as follows:

```
>>zero(GCL) <enter>
ans =
    -1.0000
    -2.0000
>>pole(GCL) <enter>
ans =
    -2.0000
    -1.3584+1.3634i
    -1.3584-1.3634i
     0.1132+1.9000i
     0.1132-1.9000i
     1.4904
```

Due to the presence of three poles with positive real parts, the closed-loop system is *unstable*.

Example 2.28 indicates that, to determine the performance, stability, and robustness of a multivariable, linear, time-invariant control system, we should be first able to express its transfer matrix $G(s)$, as a matrix of rational transfer functions, $G_{ij}(s)$, given by

$$G(s) = \begin{bmatrix} G_{11}(s) & G_{12}(s) & \dots & G_{1m}(s) \\ \dots & \dots & \dots & \dots \\ G_{n1}(s) & G_{n2}(s) & \dots & G_{nm}(s) \end{bmatrix} \quad (2.197)$$

where n is the number of outputs and m is the number of inputs. Example 2.28 also illustrates that we can extend the logic of *poles* and *zeros* of a single-input, single-output system to multivariable systems by defining the roots of the denominator and numerator polynomials of all the elemental transfer functions, $G_{ij}(s)$, as the system's poles and zeros, respectively. Such a definition also suggests that the *characteristic polynomial* of the multivariable system be defined as a single common denominator polynomial, and the roots of the characteristic polynomial are the poles of the system. Then we can apply the same stability criteria as for the single-input, single output systems by looking at the

locations of the system's poles. Similarly, the determination of responses to the singularity functions (or to an arbitrary input) can be carried out from $\mathbf{G}(s)$ in the same manner as we did for the single-input, single output systems, if we consider one input at a time. The robustness of a multivariable system can be determined using frequency response methods (see Chapter 7), provided we define scalar measures for multivariable systems analogous to gain and phase margins.

For analyzing the robustness of a general closed-loop multivariable system, consider the multivariable counterpart of the single-input, single-output closed-loop system of Figure 2.32, in which we take $\mathbf{G}(s)$ and $\mathbf{H}(s)$ to denote the plant and controller transfer matrices, respectively. Note that for the system of Figure 2.32, the closed-loop transfer matrix, $\mathbf{G}_c(s)$, defined as $\mathbf{Y}(s) = \mathbf{G}_c(s)\mathbf{Y}_d(s)$, is given by

$$\mathbf{G}_c(s) = [\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)]^{-1}\mathbf{G}(s)\mathbf{H}(s) \quad (2.198)$$

where $\mathbf{I}$ is identity matrix, $\mathbf{G}(s)$ is the plant's transfer matrix, and $\mathbf{H}(s)$ is the transfer matrix of the controller. In a manner similar to the single-loop feedback systems, we can define a *return difference matrix*. However, there are two different *return difference matrices* for the multivariable systems: the *return difference matrix at the plant's output*, $[\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)]$, and the *return difference matrix at the plant's input*, $[\mathbf{I} + \mathbf{H}(s)\mathbf{G}(s)]$. These return difference matrices are the *coefficient matrices* of the output, $\mathbf{Y}(s)$, and the input, $\mathbf{U}(s)$, respectively, when the expressions relating the desired output, $\mathbf{Y}_d(s)$, with $\mathbf{Y}(s)$ and $\mathbf{U}(s)$, respectively, are written as $[\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)]\mathbf{Y}(s) = \mathbf{G}(s)\mathbf{H}(s)\mathbf{Y}_d(s)$ and $[\mathbf{I} + \mathbf{H}(s)\mathbf{G}(s)]\mathbf{U}(s) = \mathbf{H}(s)\mathbf{Y}_d(s)$. Continuing the analogy with the single-input, single-output systems, the matrices $\mathbf{G}(s)\mathbf{H}(s)$ and $\mathbf{H}(s)\mathbf{G}(s)$ are called the *return ratio matrices* at the plant *output* and *input*, respectively. For studying robustness of the closed-loop system, the return difference (and return ratio) matrices are considered important, as we shall discuss in Chapter 7.

Before we can define quantities analogous to gain and phase margins of single-loop systems, we have to assign a *scalar* measure to the return difference matrix at the output, $[\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)]$. Such a scalar measure assigned to a matrix is its *determinant*. Since the determinant of the return difference matrix at the output is set to zero (i.e. $|\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)| = 0$) is the characteristic equation which determines the closed-loop system's poles, we obtain stability robustness information by studying Bode or Nyquist plot of the determinant of the return difference matrix at the output, $|\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)|$.

Since the return difference matrix at the output appears as an inverse in the closed-loop transfer function, we expect $[\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)]$ to be *singular* (i.e. a matrix whose inverse is not defined because its determinant is zero) at the poles of the system. If the closed-loop system is asymptotically stable, the poles must lie in the left-half s -plane. The stability robustness in this case is, therefore, a measure of how much the controller transfer matrix, $\mathbf{H}(s)$, is allowed to vary before the poles cross into the right half s -plane.

Example 2.29

Consider a two-input, two-output chemical process plant with transfer matrix $\mathbf{G}(s)$ (Figure 2.52) controlled by three flow control servo-valves, each of which has a constant transfer function (or gain), K . The outputs, $Y_1(s)$, and $Y_2(s)$, are fed back to the individual servo-control valves as shown. The plant's transfer matrix, $\mathbf{G}(s)$,

is expressed as

$$\mathbf{G}(s) = \begin{bmatrix} 3/s & -200/[s(s+6)(s+30)] \\ 0.05/s & -250/[s(s+6)(s+30)] \end{bmatrix} \quad (2.199)$$

The closed-loop system of Figure 2.57 can be reduced to that of Figure 2.55(c), with the *plant* transfer matrix, $\mathbf{G}_1(s) = \mathbf{G}(s)$, given by Eq. (2.199), and the *controller* transfer matrix, $\mathbf{G}_2(s) = \mathbf{H}(s)$, given by

$$\mathbf{H}(s) = \begin{bmatrix} K & 0 \\ 0 & K \end{bmatrix} = K\mathbf{I} \quad (2.200)$$

For a given value of controller design parameter, K , the resulting *closed-loop* transfer matrix, $\mathbf{G}_c(s) = [\mathbf{I} + \mathbf{G}_1(s)\mathbf{G}_2(s)]^{-1}\mathbf{G}_1(s) = [\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)]^{-1}\mathbf{G}(s)$, can

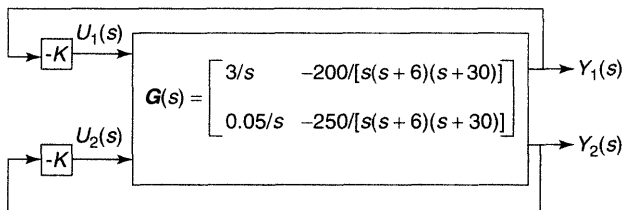


Figure 2.57 A chemical plant with transfer matrix $\mathbf{G}(s)$, controlled by two feedback servo-valves, each of transfer function, K

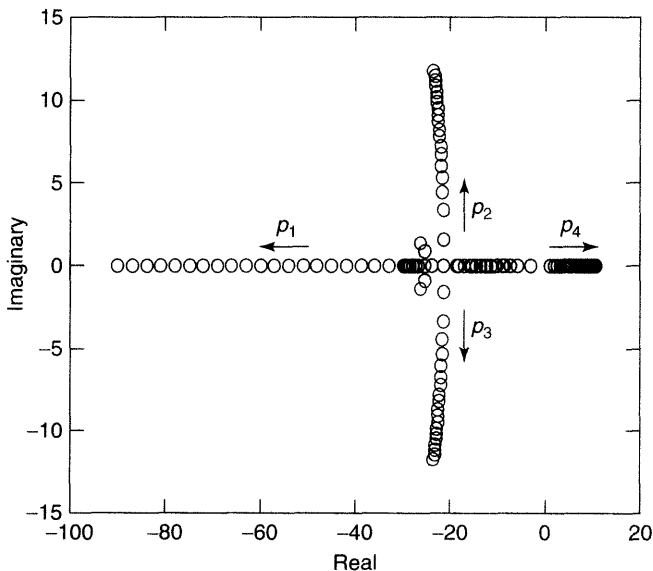


Figure 2.58 Root-locus of the multivariable chemical control system as the controller gain, K , varies from 1 to 10. The arrows indicate how each of the four poles p_1 , p_2 , p_3 , and p_4 move as K is increased

be calculated using MATLAB (CST) with the use of the *feedback* command. Alternatively, the closed-loop poles of the system are obtained by setting the determinant of the return difference matrix at the output to zero, i.e. $|\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)| = 0$, which results in the following characteristic equation:

$$|\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)| = s^4 + (36 + 3K)s^3 + (180 + 108K)s^2 + 290Ks - 740K^2 = 0 \quad (2.201)$$

Figure 2.58 is a plot of the closed-loop poles (i.e. roots of Eq. (2.201)) p_1 , p_2 , p_3 , and p_4 as the controller gain K varies from 1 to 10. It is noted that for all values of K , one of the poles (p_4) is always in the right-half s -plane, indicating instability. Hence, the system is unstable in the range $1 < K < 10$. Although for this two-loop system we can plot the root-locus by using the determinant $|\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)|$, such a procedure quickly becomes unwieldy as the number of inputs and outputs is increased.

Example 2.29 showed how $|\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)| = 0$ (i.e. the return difference matrix at the output, $[\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)]$, being singular) can be used to find out whether a *multivariable* system is stable. However, the determinant of a matrix is not a very good measure of *how close the matrix is to being singular*. Thus a Bode or Nyquist plot of $|\mathbf{I} + \mathbf{G}(s)\mathbf{H}(s)|$ in the frequency domain is *not* going to give us a very *accurate* information about the stability robustness (such as gain and phase margins), and we need another *scalar* measure which is a better indicator of how close the return difference matrix is to being singular. One such measure is the set of *singular values* of the return difference matrix in the frequency domain ($s = i\omega$), which we will study in Chapter 7.

Examples 2.28 and 2.29 illustrate the increased difficulty of estimating stability of a multivariable system using the classical frequency domain methods. While MATLAB allows that multivariable systems can be represented by their transfer matrices, an evaluation of their performance would require calculation of step (or impulse) responses of individual elements of the transfer matrix – a task which becomes unwieldy as the number of inputs and outputs are increased. A better mathematical procedure of handling arbitrary input responses of multivariable systems is to employ the methods of matrix algebra in a more tractable manner through the introduction of a *state-space* model of the system. The chief advantage of the state-space methodology is that time responses of a multivariable system are obtained directly by solving a set of first-order differential equations (called *state equations*) which are used to represent the governing differential equations. The direct calculation of a system's response to an arbitrary input enables the design and analysis of multivariable systems in the time (rather than frequency, or Laplace) domain. The state-space approach treats a single-input, single-output system in exactly the same way as it does a multivariable system, irrespective of the number of inputs and outputs involved. The remainder of this book will discuss the application of the state-space methods in control system analysis and design. However, as we will see in Chapter 7, analyzing robustness of multivariable systems requires a return to frequency domain analysis through complex function theory.